

William Steinley

GIS Forester / Forestry GIS Analyst • M.S. Forestry

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PROFESSIONAL SUMMARY

Forester and Geographic Information Systems (GIS) analyst with 8+ years in forestry, including 2+ years of GIS analysis and LiDAR processing, an M.S. in Forestry, and California Registered Professional Forester (RPF) exam eligibility. Federal timber sale and community protection experience on the Plumas National Forest, with multi-region timber sale layout and cruising across seven states. Light Detection and Ranging (LiDAR)-derived forest structure and terrain analysis in R, accelerated by AI-assisted development tools; ArcGIS Pro, Field Maps, and Survey123 deployment for field data collection. Lead author on published fire ecology research.

TECHNICAL SKILLS

Forestry and Field Operations: Timber Cruising (Fixed-Plot, Variable-Radius, Strip, 100%), Silvicultural Prescription Implementation, Cable, Helicopter, and Tractor Logging Systems, Prescribed Fire and Fuels Reduction, Vegetation Management, Forest Inventory, Crew Supervision and Quality Control, NEPA Requirement Implementation, Timber Sale Layout

GIS and Spatial Science: ArcGIS Pro 3.x, ArcGIS Online Content Publishing and Management, Esri Field Maps, Survey123, Spatial Analysis, LiDAR Processing (Canopy Height Model [CHM], Digital Terrain Model [DTM]), Canopy Cover Analysis, Geodatabase Design and Management, Geoprocessing, Cartographic Production, Coordinate Reference Systems, Slope and Terrain Modeling, Web Mapping and Cloud GIS

Programming and Data Management: R (lidR, terra, sf, tidyverse), Python (ArcPy, pandas, GeoPandas, rasterio), Statistical Analysis, Git and GitHub, AI-Assisted Development (Claude Code, Cursor), Automated Workflow Development, Technical Writing and Documentation

WORK EXPERIENCE

Principal / Consulting Forester

Mar 2026 – Present

Athena Resource Group, Gold Run, CA

- Write land management plans for private landowners, translating owner objectives into written prescriptions and treatment schedules. Part-time consulting practice with field projects in Texas.
- Perform the prescribed treatments personally: chainsaw and hand thinning, brush and invasive species control.
- Run business operations for the two-person firm: client acquisition, project scoping, and invoicing.

Forester III

Feb 2024 – Mar 2026

Jefferson Resource Company (JRC), Placerville, CA

- Supervised field crews of up to four personnel across ~8,000 acres on the Plumas National Forest Community Protection Project, overseeing and analyzing 500+ plots and catching layout, marking, and survey errors before they reached U.S. Forest Service (USFS) deliverables; interpreted and implemented National Environmental Policy Act (NEPA) requirements and silvicultural prescriptions in the field, including unit layout for helicopter yarding.
- Led the transition from Avenza Maps Pro to ArcGIS Online, ArcGIS Pro, Esri Field Maps, and Survey123, authoring two technical guides and the training materials that drove adoption; eliminated manual re-keying of field data and gave crews real-time cloud access.
- Automated cartographic production and spatial data processing in Python and ArcPy, replacing manual map assembly and standardizing output across deliverables, with AI-assisted development validated against manual quality-control checks.
- Generated CHM and DTM products from LiDAR point clouds in R (lidR), evaluating point density and comparing resampling approaches to select an appropriate grid resolution; produced slope and terrain derivatives informing yarding system selection and harvest unit delineation.
- Developed R and lidR routines for individual tree crown delineation and canopy cover estimation, proving an open-source alternative to licensed processing software.
- Published and maintained 40+ hosted feature layers and geodatabases and managed offline map areas across three project areas; crews collected stream, inventory plot, archaeological, and forest disease and dieback data against them for timber sale planning, layout, and resource surveys.
- Field liaison between JRC, USFS timber staff, and National Forest Foundation partners; supported contract administration and USFS best management practice compliance under a Registered Professional Forester.

Sunrise Pines Forestry, California and six other states

- Laid out timber sales across seven states, from Rocky Mountain mixed conifer to Gulf Coastal Plain pine, applying ground-based systems throughout and cable systems on steep terrain in California and Montana.
- Cruised timber using fixed-plot, variable-radius, strip, and 100% methods across varied forest types and ownerships, including 250+ plots personally run in eastern Oklahoma; compiled cruise data into volume summaries supporting appraisals, management plans, and harvest scheduling.
- Marked timber to silvicultural prescriptions from ponderosa pine and mixed conifer to bottomland hardwood and shortleaf pine; coordinated with landowners, consulting foresters, and logging contractors on implementation.

Fire Safe Council of Nevada County, Grass Valley, CA

- Implemented fuel reduction prescriptions under a Registered Professional Forester, running mechanical treatments with chainsaws, tow-chippers, tracked masticators, and hand tools for community wildfire protection. Three-month position; advanced to the forester role at Sunrise Pines.
- Constructed control lines to burn plan specifications and participated in prescribed fire ignition, holding, and mop-up operations.

Stephen F. Austin State University, Nacogdoches, TX

- Instructed lab sections of 30–50 undergraduate students per semester in Introduction to Forestry and Forest Biometrics, leading field trips and teaching mensuration, sampling design, and spatial analysis.
- Assisted faculty during the six-week summer field station program, hauling and staging field equipment and tracking accountability for 100+ students across exercises in timber cruising, silviculture, and forest management planning.

TimberPro Forest Service, Nacogdoches, TX

- Operated D5 bulldozer to construct control lines and suppress slopovers and spot fires during prescribed burn operations across East Texas pine plantations.
- Ignited prescribed burns using drip torches, inspected control lines via ATV, and suppressed spot fires with hand tools across multiple burn units.

SELECTED PROJECTS (GITHUB.COM/WOODSY-WILL)

- **Mohawk Valley West Slope LiDAR harvest planning (2026):** USGS 3DEP QL1 LiDAR to 3 ft DTM, CHM, and planning-slope rasters with PDAL and GDAL; 24 demonstration units on 1,621 ac; skyline screen of 8,540 corridors from 314 landings with PNW-39 payloads; 25-sheet 11×17 map series and Avenza GeoPDFs in PyQGIS; simulated BAF 20 cruise checked to FSH 2409.12.
- **Forest Inventory Analyzer (Rust, v0.2.0):** cruise compiler with Survey123 and Field Maps import, BAF expansion, TPA, basal area, QMD, stand tables, Student’s t sampling error and growth projections; CSV, JSON, and Excel input; CI-tested releases.

EDUCATION

Stephen F. Austin State University

Thesis: *Consequences of Altered Burn Regimes on Plant Community Diversity in the Pineywoods of East Texas*

Honors: Xi Sigma Pi Forestry Honor Society

Thesis research and writing: Jul 2023 – Feb 2024

Stephen F. Austin State University

Minor: Spatial Science · Concentration: Fire Management

Relevant Coursework: Forest Biometrics, GIS and Remote Sensing, Fire Ecology, Spatial Science, Silviculture

CERTIFICATIONS

- **California Registered Professional Forester (RPF) — Exam Eligible** (eligibility determined by the California Board of Forestry and Fire Protection)
- **ISA Certified Arborist — Exam Eligible** (International Society of Arboriculture)
- Qualified Timber Cruiser — U.S. Forest Service, Region 3
- Tree Risk Assessment Qualification (TRAQ) — International Society of Arboriculture
- CPR Training — Current
- Valid California Driver’s License — Clean record

PUBLICATIONS

Steinley, W. J., Oswald, B. P., Kidd, K. R., & Glasscock, J. (2024). *Influence of altered fire return intervals on plant diversity in the East Texas Pineywoods*. Journal of Forest Research, 13(5), 529.